

Liz Zhang

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EDUCATION

University of Pennsylvania

Philadelphia, PA

M.S. in Computer and Information Technology (GPA: 3.91/4.00)

Aug 2023 – May 2025

* Relevant coursework: Algorithms, Data Structures and Software Design, Artificial Intelligence, Computer Systems Programming, Computer Systems and Architecture, Software Development and Object-Oriented Programming, Big Data Analytics, Database and Information Systems

Beijing Sport University

Beijing, China

B.A. in Journalism (GPA: 3.94/4.00, ranked 15/120)

Sep 2015 – Jul 2019

SKILLS

Languages: Java, HTML/CSS/JavaScript, TypeScript, SQL, Python, C++, C, L^AT_EX

Frameworks: React, Django, Node.js/Express.js, Apache Hadoop/Spark

Tools: Git, MySQL, Postgres, NoSQL (MongoDB, Neo4j), Unix/Linux, bash, Docker, REST Api, AWS EC2/S3/RDS/EMR, Scikit-Learn, PyTorch, TensorFlow, Data Vault, dbt, Snowflake

EXPERIENCE

Software Engineer Intern

Remote

SkyIT Services

Jan 2025 – Present

- Develop and optimize 10+ reusable **React** components using **TypeScript, JavaScript, and CSS Modules** for a SaaS-based web application, reducing frontend re-rendering time by 25% and improving load time
- Resolve 20+ functionality bugs across **frontend and backend**, enhancing system stability and reducing production issues
- Implement **RESTful API** endpoints using **Node.js/Express** and **PostgreSQL**, supporting 10,000+ daily API requests with a 20% improvement in response time and reliability
- Refactor large-scale legacy code across multiple modules with **Python (Django)** and **Node.js**, improving code readability, maintainability, and test coverage
- Optimize the **CI/CD pipeline** using GitHub Actions, cutting deployment time by 50% and minimizing manual intervention
- Integrate **AWS S3** for file storage, reducing data retrieval latency by 35% and optimizing storage operations
- Automate unit and integration tests with **Jest** and **Pytest**, improving test coverage from 60% to 85%
- Deliver weekly progress reports in Agile/Scrum meetings, improving sprint planning accuracy by 20% and ensuring feature delivery timelines were met 95% of the time

Backend Engineer Intern

Remote

Peet's Coffee

Aug 2024 – Sep 2024

- Developed and deployed scalable data pipelines using **dbt, Python, and SQL**, processing millions of records daily for real-time web-based analytics
- Optimized **SQL queries** and data workflows, reducing pipeline runtime by 30% and cutting resource consumption by 20%. Achieved hourly data synchronization by refactoring key **ETL** processes
- Designed data quality checks and alerting mechanisms using dbt tests and Python scripts, increasing system stability and reducing data discrepancies by 40%
- Wrote comprehensive technical documentation for data pipeline architecture, validation workflows, and performance optimization strategies to facilitate knowledge sharing within the team

Teaching Assistant

Philadelphia, PA

University of Pennsylvania

Jun 2024 – Jul 2024, Sep 2024 – Dec 2024

- Instructed 100+ students on **Java** and **Python** programming concepts through lectures and project-based learning, focusing on **object-oriented programming, data structures**, and Python libraries such as **pandas, NumPy, and matplotlib**, with 90% of students successfully completing advanced-level projects
- Debugged and reviewed 200+ Java and Python projects, providing detailed feedback that helped significantly reduce common programming errors and improve code quality

RELEVANT PROJECTS

Research Paper Search Tool *React.js, Express.js, MySQL, AWS RDS, DataGrip*

Jan 2024 – May 2024

- Designed and developed a full-stack web application using **React.js** and **Express.js**, enabling students to search through 200M+ scientific articles
- Leveraged **Apache Spark** on Databricks to preprocess datasets, improving query speed by 35%
- Managed databases using **MySQL, AWS RDS, and DataGrip**, ensuring 99.9% uptime and efficient data storage

Local Hotspot Discovery App *React.js, JavaScript, CSS, HTML, MongoDB, Node.js, Express.js*

Sep 2024 – Dec 2024

- Engineered a dynamic app using **React.js, Node.js, Express.js, and MongoDB**, enabling 100+ users to share and explore local hotspots through images and locations
- Integrated Google Maps API for real-time location sharing and personalized recommendations, enhancing user experience and engagement